

# **NET+ 009 labs listing by module**

## **1.0 Explaining Network Topologies**

- 1.1 Lab: Create Network Topologies
- 1.2 Lab: Explore a Single Location in a Lab
- 1.3 Lab: Create a Home Wireless Network
- 1.4 Lab: Create a SOHO Network
- 1.5 Lab: Troubleshooting Methodology

## **2.0 Supporting Cabling and Physical Installations**

- 2.1 Lab: Reconnect to an Ethernet Network
- 2.2 Lab: Connect to an Ethernet Network
- 2.3 Lab: Connect a Cable Modem
- 2.4 Lab: Explore Multiple Locations in a Lab
- 2.5 Lab: Connect Network Devices
- 2.6 Lab: Connect Patch Panel Cables 1
- 2.7 Lab: Connect Patch Panel Cables 2
- 2.8 Lab: Connect Fiber Optic Cables
- 2.9 Lab: Explore Physical Connectivity 1
- 2.10 Lab: Explore Physical Connectivity 2
- 2.11 Lab: Troubleshoot Physical Connectivity 1
- 2.12 Lab: Troubleshoot Physical Connectivity 2
- 2.13 Lab: Troubleshoot Physical Connectivity 3
- 2.14 Lab: Troubleshoot Physical Connectivity 4

## **3.0 Configuring Interfaces and Switches**

- 3.1 Lab: Select and Install a Network Adapter
- 3.2 Lab: Connect a Media Converter
- 3.3 Lab: Install a Switch in the Rack
- 3.4 Lab: Secure a Switch
- 3.5 Lab: Cisco IOS Basics

- 3.6 Lab: Configure Port Aggregation
- 3.7 Lab: Enable Jumbo Frame Support
- 3.8 Lab: Configure PoE
- 3.9 Lab: Troubleshoot Disabled Ports
- 3.10 Lab: Switching Loop

## **4.0 Configuring Network Addressing**

- 4.1 Lab: Explore Packets and Frames
- 4.2 Lab: Explore ARP in Wireshark
- 4.3 Lab: Configure IP Addresses
- 4.4 Lab: Configure IP Addresses on Mobile Devices
- 4.5 Lab: Configure IP Addresses on Linux
- 4.6 Lab: Configure IP Networks and Subnets
- 4.7 Lab: IPv4 Troubleshooting Tools
- 4.8 Lab: IPv4 Troubleshooting tools for Linux
- 4.9 Lab: Use IPv4 Test Tools
- 4.10 Lab: Configure an IPv6 Address
- 4.11 Lab: Use ping and tracert on Windows
- 4.12 Lab: Use ping and traceroute on Linux
- 4.13 Lab: Assisted Troubleshooting 1
- 4.14 Lab: Assisted Troubleshooting 2
- 4.15 Lab: Assisted Troubleshooting 3
- 4.16 Live Lab: Explore the VM Lab Environment
- 4.17 Applied Live Lab: Troubleshoot IP Configuration

## **5.0 Configuring Routing and Advanced Switching**

- 5.1 Lab: Install an Enterprise Router
- 5.2 Lab: Cisco Troubleshooting Tools
- 5.3 Lab: Configure NAT
- 5.4 Lab: Create a Three-Tier Network
- 5.5 Lab: Configure Switch IP and VLAN - GUI
- 5.6 Lab: Create VLANs - GUI
- 5.7 Lab: Configure Trunking
- 5.8 Lab: Configure Switch IP Settings - CLI
- 5.9 Lab: Configure Management VLAN Settings - CLI

## 6.0 Implementing Network Services

- 6.1 Lab: Explore Three-Way Handshake in Wireshark
- 6.2 Lab: View Open Ports with netstat
- 6.3 Lab: Configure a DHCP Server
- 6.4 Lab: Configure DHCP Server Options
- 6.5 Lab: Create DHCP Exclusions
- 6.6 Lab: Create DHCP Client Reservations
- 6.7 Lab: Configure Client Addressing for DHCP
- 6.8 Lab: Explore APIPA Addressing
- 6.9 Lab: Explore APIPA Addressing in Network Modeler
- 6.10 Lab: Configure a DHCP Relay Agent
- 6.11 Lab: Add a DHCP Server on Another Subnet
- 6.12 Lab: Troubleshoot Address Pool Exhaustion
- 6.13 Applied Live Lab: Troubleshoot Address Pool Exhaustion
- 6.14 Lab: Explore DHCP Troubleshooting
- 6.15 Lab: Troubleshoot IP Configuration 1
- 6.16 Lab: Troubleshoot IP Configuration 2
- 6.17 Lab: Troubleshoot IP Configuration 3
- 6.18 Lab: Configure DNS Addresses
- 6.19 Lab: Create Standard DNS Zones
- 6.20 Lab: Create Host Records
- 6.21 Lab: Create CNAME Records
- 6.22 Lab: Troubleshoot DNS Records
- 6.23 Applied Live Lab: Configure DNS Records
- 6.24 Lab: Explore nslookup
- 6.25 Lab: Use nslookup
- 6.26 Applied Live Lab: Report DNS Configuration

## 7.0 Explaining Application Services

- 7.1 Lab: Configure NTP on Linux
- 7.2 Applied Live Lab: Troubleshoot Time Synchronization Issues
- 7.3 Live Lab: Verify Secure Web Services
- 7.4 Lab: Scan for Web Services with Nmap
- 7.5 Lab: Connect VoIP 1
- 7.6 Lab: Connect VoIP 2

7.7 Lab: Configure NIC Teaming

7.8 Live Lab: Configure First Hop Redundancy

## **8.0 Supporting Network Management**

8.1 Live Lab: Backup and Restore Network Appliances

8.2 Lab: Update Firmware

8.3 Live Lab: Update Network Documentation

8.4 Lab: Scan Using Zenmap

8.5 Applied Live Lab: Perform Network Discovery

8.6 Applied Live Lab: Configure SNMP

8.7 Lab: Configure Logging in pfSense

8.8 Lab: Evaluate Event Logs in pfSense

8.9 Lab: Auditing Device Logs on a Cisco Switch

8.10 Lab: Configure Logging on Linux

8.11 Lab: View Event Logs

8.12 Live Lab: Configure Log Collection

8.13 Lab: Troubleshoot with Wireshark

8.14 Lab: Configure Port Mirroring

8.15 Lab: Configure QoS

8.16 Live Lab: Configure Flow Collection and Analysis

8.17 Applied Live Lab: Troubleshoot Network Service Issues

## **9.0 Explaining Network Security Concepts**

9.1 Lab: Create a Honeypot

9.2 Lab: Analyze a DoS Attack

9.3 Lab: Analyze a DDoS Attack

9.4 Lab: Poison ARP and Analyze with Wireshark

9.5 Lab: Spoof MAC Addresses with SMAC

9.6 Lab: Perform a DHCP Spoofing On-Path Attack

9.7 Lab: Discover a Rogue DHCP Server

9.8 Lab: Configure DHCP Snooping

9.9 Lab: Poison DNS

9.10 Lab: Analyze DNS Spoofing

9.11 Applied Live Lab: Analyze Network Attacks

9.12 Lab: Respond to Social Engineering Exploits

9.13 Lab: Crack a Password with John the Ripper

## **10.0 Applying Network Security Features**

- 10.1 Live Lab: Deploy a Digital Certificate
- 10.2 Lab: Manage Account Policies
- 10.3 Live Lab: Configure Management Privileges
- 10.4 View Linux Services
- 10.5 Lab: Scan for Unsecure Protocols
- 10.6 Lab: Enable and Disable Linux Services
- 10.7 Lab: Disable Network Service
- 10.8 Lab: Secure Access to a Switch
- 10.9 Lab: Secure Access to a Switch 2
- 10.10 Lab: Disable Switch Ports - GUI
- 10.11 Lab: Harden a Switch
- 10.12 Lab: Configure Network Security Appliance Access
- 10.13 Lab: Configure a Security Appliance
- 10.14 Lab: Configure a Perimeter Firewall
- 10.15 Lab: Restrict Telnet and SSH Access
- 10.16 Lab: Permit Traffic
- 10.17 Lab: Block Source Hosts
- 10.18 Applied Live Lab: Troubleshoot Service and Security Issues

## **11.0 Supporting Network Security Design**

- 11.1 Lab: Configure a Screened Subnet (DMZ)
- 11.2 Lab: Configure Screened Subnets
- 11.3 Lab: Implement Intrusion Prevention
- 11.4 Lab: Scan for IoT Devices
- 11.5 Lab: Implement Physical Security

## **12.0 Configuring Wireless Networks**

- 12.1 Lab: Configure Wireless Profiles
- 12.2 Lab: Design an Indoor Wireless Network
- 12.3 Lab: Design an Outdoor Wireless Network
- 12.4 Lab: Implement an Enterprise Wireless Network
- 12.5 Lab: Configure a Captive Portal
- 12.6 Lab: Create a Guest Network for BYOD
- 12.7 Lab: Secure an Enterprise Wireless Network

- 12.8 Lab: Secure a Home Wireless Network
- 12.9 Lab: Enable Wireless Intrusion Prevention
- 12.10 Lab: Explore Wireless Network Problems
- 12.11 Lab: Troubleshoot Wireless Network Problems
- 12.12 Lab: Optimize a Wireless Network

## **13.0 Comparing Remote Access Methods**

- 13.1 Lab: Configure a Remote Access VPN
- 13.2 Lab: Configure an iPad VPN Connection
- 13.3 Lab: Configure a RADIUS Solution
- 13.4 Lab: Allow Remote Desktop Connections
- 13.5 Lab: Use PowerShell Remote
- 13.6 Live Lab: Configure a Jump Box

## **14.0 Summarizing Cloud Concepts**

- 14.1 Lab: Configure an iSCSI Target
- 14.2 Lab: Configure an iSCSI Initiator
- 14.3 Live Lab: Deploy a Cloud VM
- 14.4 Live Lab: Configure Cloud Networking

## **A.0 Network Sandbox**

- A.1 Network Sandbox Lab